

XINYANG, China

WMO: 572970

Lat: 32.1372N

Lon: 114.0433E

Elev: 115

StdP: 99.95

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.4	-3.1	-15.0	1.0	2.9	-12.1	1.3	3.3	7.7	5.3	6.8	4.1	1.8	340	0.385

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.4	35.0	26.6	33.8	25.9	32.6	25.3	28.0	32.6	27.4	31.9	26.8	31.2	3.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.8	22.7	30.5	26.2	21.9	30.2	25.6	21.2	29.7	90.8	32.5	88.2	32.1	85.5	31.2	30.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.4	6.3	5.4	-6.9	37.1	1.9	1.2	-8.3	37.9	-9.4	38.6	-10.4	39.3	-11.8	40.1		
			-8.0	28.9	1.7	0.7	-9.2	29.3	-10.2	29.7	-11.1	30.1	-12.3	30.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.3	2.8	5.6	11.3	17.1	22.0	25.6	28.0	26.9	22.5	17.3	11.0	5.2
	DBStd	9.33	3.49	4.33	4.74	4.41	3.39	2.78	2.81	3.14	3.24	3.53	4.26	3.57
	HDD10.0	592	223	134	41	3	0	0	0	0	0	1	37	153
	HDD18.3	1837	480	357	223	72	8	1	0	1	5	61	223	407
	CDD10.0	2905	2	11	81	218	372	467	559	524	374	228	66	5
	CDD18.3	1108	0	0	5	37	121	218	301	266	128	31	2	0
	CDH23.3	10685	0	1	29	273	952	2072	3468	2797	909	178	8	0
	CDH26.7	4247	0	0	4	61	296	790	1604	1180	282	31	0	0
Wind	WSAvg	2.4	2.2	2.4	2.9	2.8	2.5	2.4	2.5	2.4	2.2	2.0	2.2	2.2
Precipitation	PrecAvg	1115	29	45	67	94	121	136	208	158	113	71	53	21
	PrecMax	1567	97	107	157	308	253	309	536	487	349	232	157	68
	PrecMin	619	0	0	8	12	24	16	26	29	13	0	0	0
	PrecStd	251	21	31	37	70	56	88	133	114	70	56	42	20
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.2	20.4	26.6	30.6	33.2	35.1	36.9	37.1	33.9	29.9	24.6	17.8
		MCWB	8.2	12.7	16.4	20.3	21.6	24.1	27.2	26.6	24.3	19.0	15.2	9.6
	2%	DB	12.8	17.3	23.4	28.5	31.4	33.4	35.4	34.7	31.6	27.3	22.0	15.1
		MCWB	6.5	10.3	14.6	18.8	21.0	23.9	27.1	27.0	23.1	18.2	13.9	8.0
	5%	DB	10.6	14.7	21.1	26.5	29.9	32.2	34.1	33.3	29.8	25.4	19.7	13.0
		MCWB	5.1	8.5	13.5	17.8	20.8	23.8	26.8	26.6	22.4	16.9	12.6	6.8
	10%	DB	8.6	12.5	18.9	24.5	28.2	30.8	32.9	31.8	28.1	23.4	17.6	10.9
		MCWB	4.2	7.2	12.2	16.7	20.0	23.5	26.4	26.0	21.6	16.6	11.8	6.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.7	14.4	18.5	22.4	24.7	27.2	28.7	28.7	26.8	21.8	16.9	10.5
		MCDB	14.2	18.9	23.2	27.3	28.8	32.0	33.6	32.8	31.8	24.5	20.9	14.9
	2%	WB	7.2	11.1	16.2	20.7	23.5	26.3	28.0	28.0	25.5	19.9	15.1	9.3
		MCDB	11.5	15.2	21.6	25.7	28.0	30.7	32.9	32.4	29.2	24.1	19.8	13.2
	5%	WB	5.8	9.3	14.5	19.1	22.3	25.5	27.5	27.5	24.2	18.9	13.8	7.9
		MCDB	9.7	13.2	19.8	24.5	27.0	29.6	32.3	31.6	27.4	22.9	18.1	11.8
	10%	WB	4.6	8.0	12.9	17.7	21.3	24.7	27.0	26.8	23.0	17.9	12.6	6.7
		MCDB	7.9	11.4	17.7	22.9	26.0	28.6	31.8	30.6	26.2	21.9	16.5	10.0
Mean Daily Temperature Range	5% DB	MDBR	6.9	7.5	8.1	8.6	8.3	7.5	6.4	6.6	7.4	8.0	7.9	7.2
		MCDBR	11.1	11.5	11.2	11.1	10.7	9.6	8.1	8.5	10.1	11.5	11.4	11.2
		MCWBR	6.0	6.1	5.1	4.9	4.1	3.3	2.7	2.8	3.4	4.0	4.7	5.6
	5% WB	MCDBR	9.7	9.5	10.0	9.4	8.5	7.5	7.3	7.5	7.8	8.9	9.3	9.2
		MCWBR	5.7	5.6	5.5	5.4	4.3	3.4	3.0	2.9	3.8	4.0	4.6	5.2
Clear-Sky Solar Irradiance	taub	0.631	0.689	0.699	0.692	0.689	0.742	0.699	0.724	0.709	0.701	0.666	0.622	
	taud	1.560	1.495	1.499	1.553	1.598	1.536	1.651	1.589	1.583	1.537	1.558	1.601	
	Ebn at Noon	554	576	620	654	663	626	651	625	606	559	523	532	
	Edh at Noon	227	265	282	277	267	283	251	264	256	250	225	207	
All-Sky Solar Radiation	RadAvg	2.13	2.49	3.67	4.35	4.60	4.64	4.61	4.31	3.63	2.98	2.43	2.08	
	RadStd	0.32	0.55	0.35	0.43	0.40	0.44	0.58	0.58	0.51	0.41	0.35	0.37	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.31	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+92	N/A
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page